## Direct and inverse variation

**1** $k = 2$

**2**

**a** $M = kN^2$

**b** $k = 4$

**3**

**a** $y = kx^3$

**b** $k = 3$

**4**

**a** $y = kx^n$

**b** $k = 8$

**5**

**a** $y = kx^n$

**b** $k = -6$

**6** $k = 2$

**7** $k = \dfrac{3}{4}$

**8**

**a** $m = \frac{1}{4}$

**b** $k = 2$

## Inverse variation

**9** $y = \dfrac{k}{x}$

**10** $k = 150$

**11**

**a** Direct

**b** Inverse

**12** $r = \dfrac{k}{a}$

**13**

**a** No

**b** No

**c** No

**d** Yes

**e** Yes

**f** No

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**14**

**a** $375$ **b** $\frac{125}{2}$ **c** $\frac{125}{4}$

**15**

**a** Yes **b** Yes **c** No **d** No

**16**

**a** $k = 252$ **b** $x = 9, y = 42$

**17**

**a** $t = \frac{48}{s}$ **b** $t = \frac{2}{s^2}$

**18**

**a** $0$ **b** $0$

## Applications

**19**

**a** $1.27$ AUD/USD **b** $0.79$ USD/AUD

**20**

**a** $H = kM$ **b** $k = 0.006$ **c** $H = 0.342$ **d** $M = 14$

**21**

**a** $C = k\sqrt[4]{B}$ **b** $k = 17.4$ **c** $C = 43.8$ **d** $B = 2.6$

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